

SuperOS-303 Combined Edition v0.9.9

What's new since v0.9.7 — user guide for the **combined SuperOS+D650C** firmware (`combined650csuperOS` branch, July 2026). For experimental users.

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Terminology. **FN** = the FUNCTION key. **TAP** = the WRITE/NEXT/TAP key. "Pattern Write"/"Pattern Play" are mode-dial positions. Note keys are named **C** ... **C2** like the keyboard legend; **C** ... **C2** white keys double as pattern keys 1–8.

1 Two firmwares in one box

The unit now holds **two complete operating systems** and boots one of them at a time:

- **SuperOS-303** — the modern OS: 64-step patterns, variations, polyphony, scales, USB MIDI, web editor.
- **D650C** — a bit-perfect emulation of the original TB-303 CPU running the genuine Roland mask ROM. It behaves exactly like a stock 1982 TB-303, including the original write workflow and quirks.

Switching firmware

1. Open the config menu: hold **FN** and press **CLEAR** (works in both firmwares).
2. Look at the **G#** LED:  **solid** = **SuperOS is running**, **blinking** = **D650C is running**.
3. Press **G#**. The unit saves everything, waits until all buttons are released, and reboots into the other firmware in about a second.

Both memories are kept. SuperOS patterns live in flash; D650C patterns live in a separate memory (see section 2). Switching back and forth never erases either side.

Keep your hands off the buttons while it swaps. The firmware waits for a quiet panel before rebooting, so a held key simply delays the swap by a moment. If the unit ever appears stuck, power-cycle it; the selection is already saved and it will boot into the chosen firmware.

2 D650C mode: the original TB-303

In D650C mode the unit is a stock TB-303: use the original owner's-manual workflow (pattern clear, PITCH mode, TIME mode with the tap metronome, tracks, chaining, transpose). Nothing from SuperOS applies here except the additions below.

Pattern memory

The original battery-backed pattern RAM is replaced by permanent on-chip storage. Patterns save **automatically in the background** (a second or two after you stop editing/playing) with zero playback stutter. No battery, nothing to do.

Config menu (hold **FN, press **CLEAR**)**

Control	Function
1 – 8 (+ D#)	MIDI channel 1–8; tap D# first (LED lit) for channels 9–16.
TIME	Cycle clock source: INTERNAL → MIDI clock → DIN sync. TIME LED lit = external source; PITCH LED also lit = DIN sync.
ACCENT	Toggle MIDI soft-thru.
A# + UP / DOWN	LED brightness 1–8.
G#	Switch to SuperOS (blinking = D650C is the one running now).
CLEAR or FN	Exit the menu.

MIDI in D650C mode

- **Note out:** the sequencer is mirrored to MIDI (DIN + USB) with slide legato and accent as velocity 127.
- **Note in:** while stopped, incoming notes play the 303 voice live.
- **Clock:** slaves to MIDI clock or DIN sync (select in the menu); with the internal clock it sends MIDI clock out.
- External transport (Start/Stop or the DIN run line) starts and stops the sequencer exactly like the original's sync jack.

3 64-step patterns (SuperOS, was 32)

Patterns now run up to **64 steps** (24 in triplet mode). The length editor works while holding **FN** in Pattern Write:

While holding FN	Effect
black key C# / D# / F# / G#	Set the 8-step base (8/16/24/32).
white key C ... C2	Add 1–8 steps on top of the base.
A#	Toggle the extended range (bases become 40–64).
BACK / TAP	Length –1 / +1.
DOWN	Tap-to-count: first tap sets length 1, every further tap adds a step.
UP	Toggle triplet mode (length clamps to 24).

4 New pattern storage

SuperOS patterns moved from the tiny hardware EEPROM to a large wear-levelled area of chip flash:

- Saves happen on transport stop and ~2s after you stop editing; no manual save step.
- Patterns, tracks, and settings **survive firmware updates** done over MIDI (see section 14).
- Every pattern slot stores all three variations and, where enabled, the variation-3 chord data.

5 Three variations per pattern

Every pattern slot now holds **three independent sequences** that play simultaneously:

Variation	Output	MIDI channel
1	The 303 analog voice + MIDI	main channel
2	MIDI only	own channel (default 2)
3	MIDI only, mono or polyphonic	own channel (default 3)

Each variation has its own notes, length, direction, transpose, and scale, so one pattern slot can drive a 303 line plus two external synth lines in lockstep.

Picking which variation you edit

1. Mode dial to **Pattern Write**, normal mode (no FN/PITCH/TIME held).
2. Hold **TAP**. The picker appears on **C**/**D**/**E**: the selected variation **flashes** , the other two show **dim** .
3. Still holding **TAP**, press **C**, **D** or **E** to select variation 1, 2, or 3.

All editing (pitch/time modes, step-select, CLEAR combos) then applies to the selected variation.
The selection returns to variation 1 when you change patterns.

Setting the variation MIDI channels

In the config menu (**FN** + **CLEAR**): hold **PITCH** while pressing a channel key to set the variation-2 channel, or hold **SLIDE** for variation 3. Without a modifier the keys set the main channel as before.

6 Variation 3 polyphony

Variation 3 can play **chords of up to four notes per step** over MIDI.

Turning poly on/off (per pattern slot)

1. Stop the sequencer.
2. Hold **TAP** (variation picker) and select variation 3 with **E**.
3. Still holding **TAP**, press **SLIDE**. The SLIDE LED shows the state:  **lit = polyphonic**,  **off = monophonic**.

Writing chords

1. With variation 3 (poly) selected as the edit target, stop the sequencer and press **PITCH** (Pattern Write) to open the chord editor.
2. Note keys **toggle** chord notes on the current step; each press auditions the chord over MIDI.
3. Tap **UP**/**DOWN** to set the octave new notes land in (it latches; tap UP twice from centre for the top octave).
4. **TIME** cycles the step between note / tie / rest. **ACCENT** and **SLIDE** set per-step flags.
5. **TAP** / **BACK** move to the next / previous step (with audition).

7 Per-pattern scales

Each pattern can carry a **scale filter**: notes are pulled to the nearest allowed pitch at playback (non-destructive; your stored notes are untouched). Random/mutate operations also respect it.

1. Pattern Write, then hold **FN** and press **ACCENT** to enter scale mode.

2. Note keys toggle the twelve pitch classes (lit = allowed). The high **C2** key toggles the scale on/off for this pattern.
 3. **Presets:** hold **FN** and press a note key to load a preset rooted on that note; pressing the same note again cycles through the 35 preset scales (major, minors, modes, pentatonics, blues, and more).
 4. Tap **FN** (press and release with no note) to exit.
-

8 Keyboard play mode

Play the 303's analog voice live from the panel keys, running or stopped:

1. Mode dial to **Pattern Play**, then hold **FN** and press **PITCH** to toggle keyboard mode.
 2. Note keys play the voice; overlapping presses slide (legato), and releasing slides back to a still-held key, like a mono synth.
 3. **Octaves:** the **TIME** key toggles octave latch (TIME LED lit = latched, the default). Latched: *tap* **DOWN** / **UP** to step through octaves 0–3 and it stays. Unlatched: *hold* **DOWN** / **UP** like the stock transpose.
 4. Everything you play also goes out over MIDI per key, so a DAW records true note lengths.
 5. Press **FN** to leave the mode.
-

9 Step-select editing

Jump straight to any step of a pattern and edit it in place:

1. Pattern Write, hold **FN**, press **PITCH** to enter step-select.
 2. Black keys pick the 8-step bank (**A#** switches to the extended banks for steps 33–64); white keys pick the step within the bank.
 3. Press **TAP** to open the detail editor for the selected step: change pitch, octave, accent and slide with audition; the playhead is not disturbed.
 4. Or press **TIME** for the time sub-mode: **DOWN** / **UP** / **ACCENT** write note / tie / rest directly on the selected step.
 5. **FN** exits. (With a chain active, **DOWN** / **UP** / **ACCENT** / **SLIDE** switch which chained pattern you are viewing.)
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10 Metronome tap-write

Write a pattern's rhythm by tapping along to a metronome, exactly like the original 303's TAP timing method, with the original's sound: a short **E5 click on every 8th note** and a lower **E4 click marking the start of each bar**.

1. Run the sequencer in **Pattern Write** (normal mode).

2. Hold **CLEAR** and press **TIME**. The pattern's rhythm is cleared and the metronome starts clicking.
3. Hold **TAP** where you want notes: held at a beat = NOTE, held through = TIE, released = REST.
4. The mode exits by itself at the end of the bar once you've tapped something (or press **CLEAR** + **TIME** again).

11 Pitch calibration (read this once)

Both firmwares now use the **factory pitch standard** from the Roland service notes: the untransposed low **C** key puts exactly 1.000V on the CV jack. SuperOS and D650C play **identical pitches** for the same key, and MIDI is aligned the same way in both (low **C** key ↔ MIDI note 48 / C3).

Coming from an older SuperOS build? Older SuperOS output sat one semitone above the factory spec. If your instrument was tuned against it, everything will now sound one semitone flat until you turn the front-panel **TUNING knob up one semitone**, once. After that, both firmwares match each other and a factory-calibrated TB-303.

Bonus: when the sequencer is stopped, the selected pattern LED keeps flashing at the tempo (120 BPM idle, or the incoming clock tempo), like a real 303.

12 USB MIDI & DAW control

The unit enumerates over USB-C as a class-compliant MIDI device named **"SuperOS-303"** (no drivers). DIN and USB carry the same traffic, both directions:

- Sequencer notes out (all three variations on their channels), clock and transport in/out.
- Live note input while stopped plays the 303 voice; velocity ≥ 100 adds accent.
- **Pattern remote control:** CC0 selects the bank (0–3), CC32 the section (0 = patterns 1–8, 1 = 9–16), and a Program Change selects the pattern — matching Ableton's bank/program menus.

13 Web pattern editor

A full remote editor runs in Chrome/Edge (WebMIDI) from the repo — no install:

```
python3 -m http.server 8080 --directory tools/web-pattern-edit
# then open http://localhost:8080 and pick the SuperOS-303 MIDI port
```

- Edit all 16 patterns of a bank, all three variations, poly chords, scales, lengths, directions.
- Live playhead chase while the sequencer runs; edits sync both ways instantly.
- Use it to **back up patterns** before firmware surgery, and to restore them after.

14 Updating the firmware

Updates are one SysEx file over MIDI; no programmer needed. `combined-v0.9.9.syx` is **self-contained** (application + flash service) — any SuperOS-303 CPU with the bootloader can take it.

1. Enter the bootloader: hold **TAP** while powering on, or send SysEx `F0 7D 4A F7` to the running unit (works in both firmwares).
2. Send `combined-v0.9.9.syx` to the "SuperOS-303" MIDI port. Over USB it takes about a second, no throttling needed. Over DIN MIDI, use a SysEx tool with a delay between messages (e.g. SysEx Librarian ~50ms).
3. The unit reboots into the app by itself when the file ends.

First install on a unit coming from v0.9.7: the pattern storage layout moved, so the first boot of the combined firmware starts with empty SuperOS patterns. Back up patterns with the web editor first and restore them after. Subsequent v0.9.9+ updates keep everything.

15 Quick reference card

Action	Where	Keys
Switch firmware (SuperOS ↔ D650C)	either firmware	FN + CLEAR menu → G# (solid = SuperOS, blinking = D650C)
Config menu	either firmware	hold FN , press CLEAR
Pick edit variation	Pattern Write	hold TAP → C / D / E (flashing = selected)
Var 3 poly on/off	Pattern Write, stopped	hold TAP → E → SLIDE (lit = poly)
Chord editor (var 3 poly)	Pattern Write, stopped	PITCH ; keys toggle notes, TAP / BACK step
Scale mode	Pattern Write	FN + ACCENT ; FN + note = presets
Keyboard play	Pattern Play	FN + PITCH ; TIME = octave latch
Step-select	Pattern Write	FN + PITCH ; black = bank, white = step, TAP = detail
Length editor	Pattern Write	hold FN : black = base, white = +1–8, A# = 33–64, UP = triplets
Metronome tap-write	Pattern Write, running	CLEAR + TIME , then tap TAP
Variation MIDI channels	config menu	hold PITCH (var 2) or SLIDE (var 3) + channel key
Bootloader (firmware update)	power-on	hold TAP while switching on, or SysEx F0 7D 4A F7

SuperOS-303 Combined Edition v0.9.9 — branch `combined650csuperOS`. D650C mode runs the genuine TB-303 mask ROM via a cycle-accurate µCOM-43 core (MAME lineage, verified against a 100M-instruction differential trace). Pitch standard per Roland TB-303 Service Notes (Feb 1982), CV adjustment procedure.